Solution Design Document

UiPath

TABLE OF CONTENTS

I. Purpose 3

II. Automated process details 4

3 Runtime guide 5

3.1 Architectural structure of the Master Project 5

3.2 Master Project Runtime Details 5

3.3 Project name 1 6

3.4 Project(s) workflows 6

3.5 Packages 7

3.6 Architectural structure of the Master Project 7

4 Other Details 8

Future Improvements 8

Other Remarks 8

5 Glossary 9

1. PURPOSE

This Solution Design Document (SDD) outlines the technical architecture of the implementation of the Google Reviews Monitoring and AI Sentiment Auto-Response Automation process developed for Automation Consultant, Kaunas. The solution automates the monitoring of Google Business reviews, performs sentiment analysis, generates AI-based reply suggestions using Claude AI, and logs all activities in a structured format.

The automation eliminates manual review checking, reduces response time to under 2 hours, and ensures consistent and professional communication with customers. It also introduces structured tracking and auditability through Google Sheets logging and automated email notifications.

The solution is designed with focus on:

- Robustness – handling API errors, missing data, and retries
- Efficiency – reducing manual effort and response time
- Replicability – standardized process that can be reused
- Reusability of components – modular use of APIs (Google Places, Sheets, Claude AI)

The document serves as a reference for the RPA developer, business owner, and any future support developers responsible for maintaining or enhancing the automation.

2. AUTOMATED PROCESS DETAILS

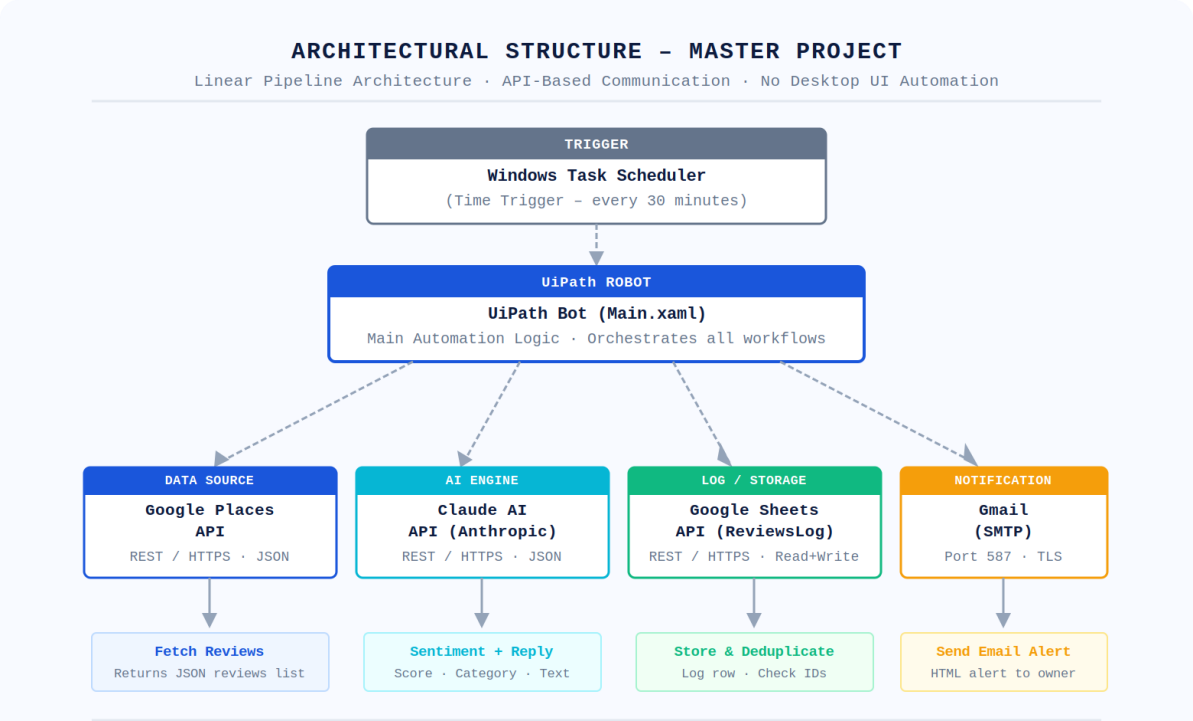
In this part main details of the automated process are described and briefly explained.

Item	Description
Master Project Name	Google Reviews Monitoring & AI Response Automation
Robot Type	Unattended
Orchestrator used?	No – scheduled via Windows Task Scheduler (every 30 min)
Scalable	Yes – can monitor multiple Google Business locations
UiPath version used	UiPath Studio 2026.0.189 (Community Edition)
Reporting	Google Sheets ReviewsLog tab (Live): https://docs.google.com/spreadsheets/d/1rrSW-JNwvnfElAsSGJ-Ks6Pwcdop-sHRE_vn-0ZdlNA/edit?usp=sharing
Stored Credentials	API keys stored in appsettings.json (not hardcoded in workflows)
Password Policies	Gmail App Password (16-char) – not expiring; Claude API key rotated annually
Schedule Details	Every 30 minutes, 24/7

3. RUNTIME GUIDE

3.1. Architectural structure of the Master Project

Looking at the schematic, the structure of the master project is explained, with main notes, as shown in the picture below.



3.2. Master Project Runtime Details

Project is explained in higher detail inside the table, as given below.

ITEM NAME	DESCRIPTION *Fill in each bolded section - empty fields are not allowed. If the section does not apply to your automation then mark as n/a.*
Trigger Mechanism	Running via UiPath Orchestrator. Scheduled every 30 min to check for new Google reviews.
Prerequisites to run	Google Cloud project with Places API enabled. Google Sheets configured and shared with service account. Valid Claude API key with credits. appsettings.json properly configured. Internet connection available.
Input Data	Google Business reviews (text, rating, author, timestamp) retrieved via Google Places API. Timestamp from Google Sheets (Config tab)
Expected output	Email sent to business owner with review details and suggested reply. Review data stored in Google Sheets (ReviewsLog tab)
How to start the automated process	Process is triggered automatically from UiPath Orchestrator based on time schedule.
Reporting (queues reporting, Kibana or another platform)	UiPath Orchestrator logs. Google Sheets used for review tracking and audit. Email notifications per review.
How is Orchestrator used?	Orchestrator used for scheduling the process every 30 min.
Password policies (mention any specific compliance requests)	Gmail App Password used for SMTP. API keys stored securely.
Stored credentials (Never use hardcoded credentials in the workflow!)	Stored in configuration files (appsettings.json) and service account JSON files.
List of queues names (Naming convention: ProcessName_QueueName)	*N/A*
Schedule Details	Every 30 min via UiPath Orchestrator trigger.
Multiple Resolutions Supported? (in case of image automation / Citrix and VDI)	*N/A*
Recommended Resolution	*N/A*

3.3. Project name

ITEM NAME	DESCRIPTION *Fill in each section - empty fields are not allowed. If the section does not apply to your automation then mark as n/a.*
Environment used for development (name, location, configuration details etc)	Local Windows machine with UiPath Studio 2026.0.189 (Community Edition).
Environment prerequisites (OS details, libraries, required apps)	Windows OS, UiPath Studio installed, Google Cloud project with Places API and Sheets API enabled, Gmail SMTP configured.
Repository for project (where is the developed project stored)	Local project folder (UiPath Studio project directory).
Configuration method (assets, excel file, Json file)	JSON configuration (appsettings.json), service account JSON files for Google APIs.
List of reused components	***N/A***
List of new reusable components	***N/A***

3.4. Project(s) workflows

These are the workflows, used in the automation.

Workflow Name	Description
Main.xaml	Main.xaml (orchestrates the process), Config loading workflow (reads appsettings.json), Review Fetch workflow (calls Google Places API), AI Processing workflow (Claude sentiment and reply), Email workflow (sends notifications), Logging workflow (writes to Google Sheets).
FetchReviews.xaml	Calls Google Places API with API key and Place ID; returns list of review objects
GetProcessedIDs.xaml	Reads Google Sheets ReviewsLog tab and returns list of already-processed review IDs
AnalyseSentiment.xaml	Calls Claude AI API with review text; returns Sentiment, Score (0-100), and SuggestedReply
LogToSheets.xaml	Writes review data + AI analysis result to Google Sheets ReviewsLog tab via Sheets API
SendEmailAlert.xaml	Sends formatted HTML email to owner via Gmail SMTP with review details and AI reply
ErrorHandler.xaml	Centralised exception handler: retries 3x, then logs to ErrorLog tab and sends error alert
Workflow Name	Description

Main.xaml	Entry point. Reads config, invokes all sub-workflows in sequence, handles global exceptions
FetchReviews.xaml	Calls Google Places API with API key and Place ID; returns list of review objects
GetProcessedIDs.xaml	Reads Google Sheets ReviewsLog tab and returns list of already-processed review IDs
AnalyseSentiment.xaml	Calls Claude AI API with review text; returns Sentiment, Score (0-100), and SuggestedReply
LogToSheets.xaml	Writes review data + AI analysis result to Google Sheets ReviewsLog tab via Sheets API
SendEmailAlert.xaml	Sends formatted HTML email to owner via Gmail SMTP with review details and AI reply
ErrorHandler.xaml	Centralised exception handler: retries 3x, then logs to ErrorLog tab and sends error alert

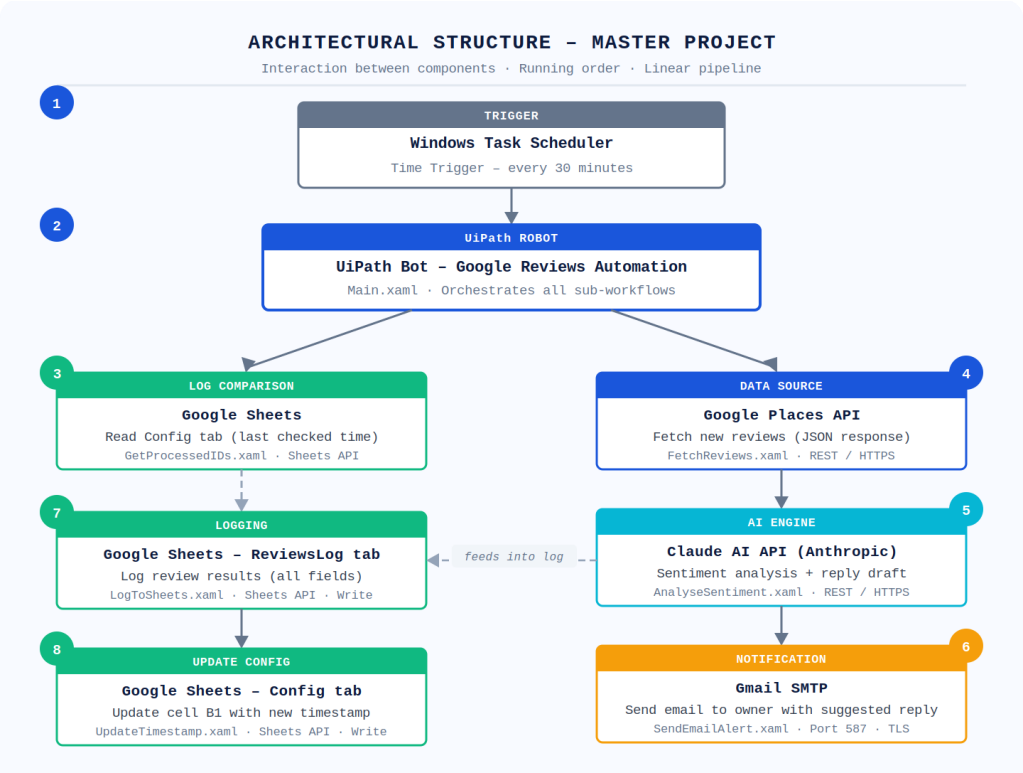
3.5. Packages

These are the packages used in the process – they are required in order to set up and run the automation prepared.

Package Name	Description
UiPath.System.Activities	File read/write, JSON serialisation, data manipulation, exception handling
UiPath.WebAPI.Activities	HTTP GET/POST requests to Google Places API, Anthropic Claude API, and Google Sheets API
UiPath.GSuite.Activities	Used to interact with Google Sheets for reading and writing review data.
UiPath.Mail.Activities	Used to send email notifications via Gmail SMTP.

3.6. Architectural structure of the Master Project

Display the interaction between components (package / robots, Orchestrator queues, and running order) in a diagram.



4. OTHER DETAILS

FUTURE IMPROVEMENTS

- Enable automatic posting of replies directly to Google after API approval
- Improve error handling and add retry mechanisms for API failures
- Enhance AI-generated responses for more personalization
- Add dashboard/reporting for review trends and sentiment analysis
- Support higher volume of reviews if business grows

Other Remarks

All reviews are stored in Google Sheets to ensure tracking and auditability. The business owner reviews and posts replies manually in the current phase. Stable internet connection and valid API credentials are required for execution.

Live Google Sheets Audit Log (ReviewsLog + ErrorLog tabs):
https://docs.google.com/spreadsheets/d/1rrSW-JNwvnfElAsSGJ-Ks6Pwcdop-sHRE_vn-0ZdlNA/edit?usp=sharing

5. Glossary

The main terms used in the Solution Architecture Document are defined below:

Master project - the overall output of the development, containing one or multiple projects that together cover the scope of the robotic process automation. There is a 1 to 1 connection between the Master Project and the Process to be automated (As presented in the PDD).

Project - an UiPath Studio project containing one or multiple workflow files. A project can be converted to a package and run independently, covering a particular scope within the master project. Or multiple projects can be converted into one package depending on the aims and restrictions of the automation. The project is used when defining the development and support phase of the automation.

Package - the output of compiling one or multiple projects. A package can be deployed on the robot machine and be executed by the robot service. Only one package can be executed at a given time by a robot. The package is used when defining the running phase of the automation.

Workflow - a component of the package, the workflow encapsulates a part of the project logic. The workflow can be of type: sequence, flowchart or state machine. A workflow is saved as an

.xaml file inside the project folder. A workflow file can be invoked from another workflow and by default there is an initial workflow file that will run when executing the package.

Activity - an action that the robot executes.

Sequence - a workflow where activities are executed one after another, in a sequential order.

Flowchart - a workflow where activities are connected by arrows and the logic of the workflow can be easily followed in a visual manner. The flowchart can also be exported as an image from UiPath studio.

State machine - a more advanced way of organizing a workflow, similar to a flowchart.

BOR - Back office robot.

FOR – Front office robot.

Orchestrator – Enterprise architecture server platform supporting: release management, centralized logging, reporting, auditing and monitoring tools, remote control, centralized scheduling, queue/robot workload management, assets management.

Google Places API – A Google service used to retrieve Google Business reviews, including review text, rating, author, and timestamp.

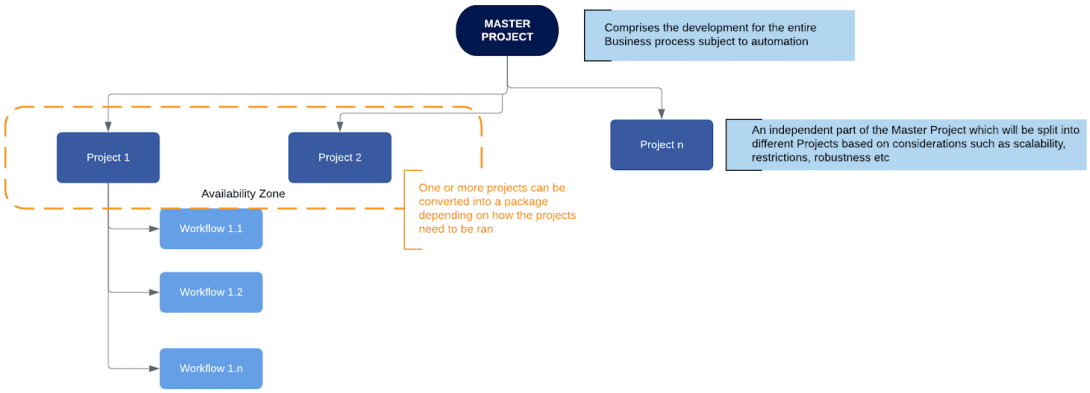
Claude AI – An AI model provided by Anthropic used in this process for sentiment analysis and generating reply suggestions.

Sentiment Analysis – The process of classifying reviews as Positive, Neutral, or Negative based on their content, along with a confidence score.

Google Sheets (ReviewsLog) – A spreadsheet used to store all processed reviews, including sentiment results and generated replies for tracking and audit purposes.

Timestamp (Checkpoint) – A stored value used to track the last processed review time, ensuring that only new reviews are processed in each run.

Email Notification – An automated email sent to the business owner containing review details and suggested replies.


MASTER PROJECT
Comprises the development for the entire Business process subject to automation
Project 1
Project 2
Project n
An independent part of the Master Project which will be split into different Projects based on considerations such as scalability, restrictions, robustness etc
Availability Zone
One or more projects can be converted into a package depending on how the projects need to be ran
Workflow 1.1
Workflow 1.2
Workflow 1.n